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Inventor(s)

PARK; Ho Sik (Hans) et al.

Component Carrier and Method of Manufacturing the Same

Abstract

A component carrier includes a stack having at least one electrically conductive layer structure and at least one inorganic layer structure, the at least one inorganic layer structure has a planar first main surface and an opposing planar second main surface. The component carrier further includes a geometrically discontinuous region at the first main surface, and a reinforcing layer structure at the second main surface. The reinforcing layer structure at least partially laterally overlaps the geometrically discontinuous region.

Inventors: PARK; Ho Sik (Hans) (Chongqing, CN), LUO; Rowling (Chongqing, CN), WANG; Tony (Chongqing, CN)

Applicant: AT & S Austria Technologie & Systemtechnik Aktiengesellschaft (Leoben, AT)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This utility patent application claims the benefit of the filing date of the patent application No. 202410274970.4, filed on Mar. 11, 2024, in the National Intellectual Property Administration of the Peoples Republic of China, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] Embodiments of the disclosure relate to a component carrier, and to a method of manufacturing a component carrier.

TECHNOLOGICAL BACKGROUND

[0003] In the context of growing product functionalities of component carriers equipped with one or more components and increasing miniaturization of such components as well as a rising number of components to be connected to the component carriers such as printed circuit boards or component carriers, increasingly more powerful array-like components or packages having several components are being employed, which have a plurality of contacts or connections, with smaller and smaller spacing between these contacts. In particular, component carriers should be mechanically robust and electrically reliable so as to be operable even under harsh conditions.

[0004] Conventional approaches of forming component carriers are still challenging.

SUMMARY

[0005] There may be a need to form a compact and reliable component carrier.

[0006] According to an exemplary embodiment of the disclosure, a component carrier is provided which comprises a stack comprising at least one electrically conductive layer structure and at least one inorganic layer structure, wherein the at least one inorganic layer structure has a planar first main surface and an opposing planar second main surface, a geometrically discontinuous region at the first main surface, and a reinforcing layer structure at the second main surface, wherein the reinforcing layer structure at least partially laterally overlaps with the geometrically discontinuous region.

[0007] According to another exemplary embodiment of the disclosure, a method of manufacturing a component carrier is provided, wherein the method comprises providing a stack comprising at least one electrically conductive layer structure and at least one inorganic layer structure, wherein the at least one inorganic layer structure has a planar first main surface and an opposing planar second main surface, forming a geometrically discontinuous region at the planar first main surface, and forming a reinforcing layer structure at the opposing planar second main surface, wherein the reinforcing layer structure at least partially laterally overlaps with the geometrically discontinuous region.

Overview of Embodiments

[0008] In the context of the present application, the term “component carrier” may particularly denote any support structure which can accommodate one or more components thereon and/or therein for providing mechanical support and/or electrical connectivity and/or thermal connectivity. In other words, a component carrier may be configured as a mechanical and/or electronic and/or thermal carrier for components. In particular, a component carrier may be one of a printed circuit board, an organic or inorganic interposer, and an IC (integrated circuit) substrate. For example, a component carrier may be a rigid-flex carrier, or a flexible substrate. A component carrier may also be a hybrid board combining different ones of the above-mentioned and/or other types of component carriers.

[0009] In the context of the present application, the term “stack” may particularly denote a flat or planar body. For instance, the stack may be a layer stack, in particular an at least partially laminated layer stack. Such a stack may be formed by connecting a plurality of layer structures, which preferably may be arranged in a parallel manner, for example by the application of mechanical pressure and/or heat. The stack may comprise at least one electrically conductive layer structure

and at least one electrically insulating structure. A stack may comprise a plurality of sub-stacks, for example may comprise a first multilayer build-up beneath and a further multilayer build-up above an inorganic layer structure.

[0010] In the context of the present application, the term “layer structure” may particularly denote a continuous layer, a patterned layer or a plurality of non-consecutive islands within a common plane. A layer structure may perform the function of providing electrical conductivity and/or electrical insulation.

[0011] In the context of the present application, the term “inorganic layer structure” may particularly denote a layer structure which comprises inorganic material, such as an inorganic compound. In particular, dielectric material of the inorganic layer structure or even the entire inorganic layer structure may be made exclusively or at least substantially exclusively from inorganic material. In another embodiment, the inorganic layer structure may comprise inorganic dielectric material and additionally another dielectric material. An inorganic compound may be a chemical compound that lacks carbon-hydrogen bonds or a chemical compound that is not an organic compound. In an example, the inorganic layer structure may comprise glass or quartz, for example silicon base glass, in particular soda lime glass, and/or borosilicate glass and/or aluminosilicate glass and/or lithium silicate glass and/or alkaline free glass. In another example, the inorganic layer structure may comprise ceramic material, for example aluminum nitride and/or aluminum oxide and/or silicon nitride and/or boron nitride and/or tungsten comprising ceramic material. Yet, in another example, the inorganic layer structure may comprise semi-conducting material, for example silicon and/or germanium and/or silicon oxide and/or germanium oxide and/or silicon carbide and/or gallium nitride. In a further embodiment, the inorganic layer structure may comprise (elemental) metal and/or metal alloys, for example, copper and/or tin and/or bronze. Yet in another embodiment, the inorganic layer structure may comprise inorganic material, which is not listed in the above-mentioned example, such as: MoS_2 , CuGaO_2 , AgAlO_2 , LiGaTe_2 , AgInSe_2 , CuFeS_2 , BeO .

[0012] In the context of the present application, the term “geometrically discontinuous region” may particularly denote any structural discontinuity at a planar main surface locally deviating from said planarity. For example, the geometrically discontinuous region may comprise an indentation (such as a cavity, a blind hole, a through hole, etc.) and/or a protrusion (such as a post, a step, a ring, an undulation structure, etc.) surrounded by a planar area of said main surface. For example, the geometrically discontinuous region may be any local non-planarity with respect to a planar surrounding area.

[0013] In the context of the present application, the term “reinforcing layer structure” may particularly denote a layer structure (in particular in accordance with the above definition) at a main surface of an inorganic layer structure, which may provide the inorganic layer structure having a geometrically discontinuous region on an opposing other main surface, with a reinforcement function. For example, the reinforcing layer structure may provide the inorganic layer structure with additional crack resistance in comparison with a scenario in which the reinforcing layer structure is absent. More specifically, the reinforcing layer structure may be configured so that the combination of the inorganic layer structure with the reinforcing layer structure is less prone to cracks than the inorganic layer structure without the reinforcing layer structure. In particular, said reinforcing layer structure may be configured for at least partially compensating stress created by the geometrically discontinuous region on the opposing main surface, for example during manufacturing the component carrier and/or during use in operation. Said reinforcing layer structure may be configured to at least partially distribute stress from the geometrically discontinuous region on the opposing main surface or the whole opposing main surface evenly to each area or corner of the reinforcing layer structure, therefore the overall stress at one area or point may be reduced significantly and this may mitigate the risk of cracks. For instance, during pressure-supported assembly of a component in a cavity constituting the

geometrically discontinuous region, stress may be applied to the opposing main surface of the inorganic layer structure, which stress may be reduced by the reinforcing layer structure. For example, during the lamination of at least one additional layer build-up on one or both opposing main surfaces of the inorganic layer structure, stress may be applied to the inorganic layer structure in view of the geometrically discontinuous region deviating from planarity. The reinforcing layer structure may be configured for at least partially compensating, absorbing and/or reducing and/or distributing stress resulting from the presence of the geometrically discontinuous region.

[0014] In the context of the present application, the term “reinforcing layer structure at least partially laterally overlapping with the geometrically discontinuous region” may particularly denote that, in a plan view of the stack in a viewing direction perpendicular to the planar main surfaces of the inorganic layer structure, the reinforcing layer structure and the geometrically discontinuous region may share at least a partial common area. For instance, there may be an overlap between reinforcing layer structure and geometrically discontinuous region in the sense that, in the plan view, a part of the reinforcing layer structure corresponds to a part of the geometrically discontinuous region, a part of the reinforcing layer structure is outside of the geometrically discontinuous region, and a part of the geometrically discontinuous region is outside of the reinforcing layer structure. Alternatively, in the plan view, the geometrically discontinuous region may lie completely inside of the reinforcing layer structure, and the reinforcing layer structure may comprise a portion outside of the geometrically discontinuous region. Further alternatively, in the plan view, the reinforcing layer structure may lie completely inside the geometrically discontinuous region, and the geometrically discontinuous region may comprise a portion outside of the reinforcing layer structure. In still another embodiment, in the plan view, the geometrically discontinuous region and the reinforcing layer structure may be in full alignment or flush with each other. The overlapping may be along the thickness or vertical direction.

[0015] In the context of the present application, the term “main surface” of a body may particularly denote one of two largest opposing surfaces of the body or outermost opposing surfaces of the body. The main surfaces may be connected by circumferential side walls. The thickness of a body, such as the inorganic layer structure, may be defined by the distance between the two opposing main surfaces.

[0016] According to an exemplary embodiment of the disclosure, a component carrier (such as a printed circuit board or an integrated circuit substrate) has a (preferably laminated) layer stack having also one or more inorganic layer structures (such as a glass plate). An inorganic layer structure may be more brittle, stiffer and/or less elastic than an organic layer structure. Said at least one inorganic layer structure may have two opposing generally planar main surfaces. However, one of said main surfaces may be provided with a geometrically discontinuous region (such as a cavity) deviating from planarity. Advantageously, the opposing main surface of the at least one inorganic layer structure may be equipped with a reinforcing layer structure in a spatial range of a horizontal plane which overlaps or corresponds to a spatial range of a horizontal plane of the geometrically discontinuous region. Due to the presence of the reinforcing layer structure in spatial relationship with the geometrically discontinuous region, damage of the inorganic layer structure during manufacturing (for instance during a panel level fabrication process) and/or during operation of the component carrier may be reliably prevented, since the reinforcing layer structure may be designed for reducing, inhibiting and/or buffering stress acting on the at least one inorganic layer structure in view of its geometrically discontinuous region. For example, the risk of glass damage during die bonding in a cavity-type geometrically discontinuous region, during which pressure is applied to the at least one inorganic layer structure, may be strongly suppressed by the reinforcing layer structure. Further advantageously, the configuration of the reinforcing layer structure as layer-type reinforcing structure may lead to a low or even no extra space consumption. Consequently, an exemplary embodiment may provide a compact and reliable component carrier.

[0017] In the following, further exemplary embodiments of the component carrier and the method

will be explained.

[0018] In an embodiment, the geometrically discontinuous region comprises a cavity extending from the first main surface into the at least one inorganic layer structure. Such a cavity may particularly denote an accommodation hole, an opening, a recess or a via in the inorganic layer structure. Such a cavity may be a through hole extending through the entire inorganic layer structure, may be a blind hole extending into the inorganic layer structure but having a closed bottom, or may be a groove or recess of any shape.

[0019] In an embodiment, the component carrier comprises at least one component, for example a bridge die, accommodated at least partially in the cavity. In the context of the present application, the term “component accommodated at least partially in the cavity” may particularly denote a component being fully accommodated or only partially accommodated in the cavity in the inorganic layer structure. In a fully accommodated embodiment, the entire vertical spatial range between an upper end and a lower end of the component is located inside of the cavity. In a partially accommodated embodiment, only part of a vertical spatial range between an upper end and a lower end of the component is located inside of the cavity, for instance the component may protrude upwardly and/or downwardly beyond the cavity. In one embodiment, the upper end of the at least partially accommodated component may be in alignment with an upper main surface of the inorganic layer structure and/or the lower end of the component may be in alignment with a lower main surface of the inorganic layer structure. For instance, the component may be an electrical component for providing an electrical function (such as a semiconductor chip), a thermal component for providing a cooling function (such as a copper or ceramic block) and/or an optical component providing an optical function. Said optical component may for instance be an optical chip, an optoelectronic device, or an optical connector. When the component is configured as a bridge die, it may be electrically coupled with a second component and with a third component for establishing an electric connection between said second component and said third component. Said second and third components may be surface mounted on and/or embedded in the inorganic layer structure.

[0020] In an embodiment, an external electrically conductive layer structure of the at least one electrically conductive layer structure is arranged on the first main surface and is arranged at another vertical level than an electrically conductive pad (such as a chip terminal) at an upper main surface of the component. Such a vertical misalignment between a component pad and an external electrically conductive layer structure on the inorganic layer structure may be the result of an embedding process.

[0021] In an embodiment, the geometrically discontinuous region comprises a protrusion extending upwardly from the first main surface beyond the at least one inorganic layer structure. Also, a protrusion at a top side of an inorganic layer structure may lead to stress on a bottom side thereof, for instance by applying stress during lamination. A reinforcing layer structure on the bottom side of the inorganic layer structure may help to prevent damage of the inorganic layer structure in view of the stress caused by the top-sided protrusion.

[0022] In an embodiment, the reinforcing layer structure is arranged on the second main surface. For example, the reinforcing layer structure may be formed as a patterned layer, for example a patterned copper foil, being applied onto the second main surface of the inorganic layer structure. This may bring the advantage of compensating and/or redistributing applied stress or pressure in an efficient way, while at the same time inhibiting crack-formation inside the inorganic layer structure. Meanwhile, the inorganic layer structure may be equipped with an electrical element to transfer electricity.

[0023] In an embodiment, the reinforcing layer structure is electrically conductive, for example forms part of the at least one electrically conductive layer structure. This may bring the advantage that substantially no extra effort is required for manufacturing the reinforcing layer structure, since it can be formed simultaneously with an electrically conductive layer structure provided in the

stack anyway. For example, a portion of a common patterned metal layer may be configured as a reinforcing layer structure, whereas another portion of said common patterned metal layer may be configured for providing an electrical function.

[0024] In an embodiment, at least a portion of the electrically conductive layer structure is formed apart from the reinforcing layer structure. For instance, the other part of the electrically conductive layer structure which is provided in addition to the reinforcing layer structure may be a peripheral part, whereas the reinforcing layer structure may be a central part of a common electrically conductive layer structure. This may bring the advantage of subdividing the electrically conductive layer structure so that it can be configured for transmitting current and/or signals. The electrically conductive layer structures may also be configured for ensuring physical and/or mechanical integrity of the stack of the component carrier.

[0025] In an embodiment, the reinforcing layer structure forms traces and/or pads, for example is a pattern which forms traces and/or pads. In such an embodiment, the reinforcing layer structure may fulfill a double function, i.e., the reinforcing layer structure may contribute to the electric functionality of the component carrier by forming traces and/or pads and may simultaneously function for reinforcing the inorganic layer structure's bottom side against stress.

[0026] In an embodiment, the reinforcing layer structure is formed as a continuous structure, for example as a continuous layer. Such a continuous structure may extend over the entire spatial area of the geometrically discontinuous region on the opposing other main surface of the inorganic layer structure. For instance, the reinforcing layer structure may be a continuous metal layer, such as a copper foil or a plating structure. This may allow the reinforcing layer structure to provide its reinforcing function over a continuous spatial range.

[0027] In an embodiment, the reinforcing layer structure extends laterally over the entire geometrically discontinuous region, for example is aligned laterally with the entire geometrically discontinuous region. This may provide a particularly reliable protection of the inorganic layer structure against damage. It may however also be that the reinforcing layer structure (partially or completely) laterally extends inside the geometrically discontinuous region.

[0028] In an embodiment, the reinforcing layer structure extends laterally over at least part of a boundary profile of the geometrically discontinuous region. This may bring the advantage of efficiently guiding and/or distributing applied stress or pressure over the portion of the stack spatially exceeding the profile of the geometrically discontinuous region resulting in a reliable crack inhibition of the inorganic layer structure. For example, said boundary profile may be formed by an edge and/or a corner portion of the geometrically discontinuous region.

[0029] In an embodiment, the reinforcing layer structure can extend over the whole opposing main surface facing away from the geometrically discontinuous region. A pattern extending beyond the area of the geometrically discontinuous region can be used to perform the function of the reinforcing layer structure or supporting structure, for example except an electrical function. With such kind of structure, the stress can be distributed to more points or areas, each single area may just have to withstand less pressure, and the entire area may withstand a higher pressure.

Meanwhile, the pattern of the whole surface can be any pattern contributing to fulfillment of the function required in accordance with a design.

[0030] In an embodiment, the reinforcing layer structure is inorganic and/or electrically insulating, for example forms part of the at least one inorganic layer structure. For example, a bottom portion of the inorganic layer structure can be constructed, structured, or patterned to form a bottom-sided profile corresponding spatially at least partially with the geometrically discontinuous region for self-protecting the inorganic layer structure against stress. The profile may be of such kind that particularly the portions on the lower main surface of the inorganic layer structure being prone to damage in the event of stress applied to the opposing geometrically discontinuous region are locally reinforced.

[0031] In an embodiment, the reinforcing layer structure is configured and connected with the stack

for dissipating heat. For instance, the reinforcing layer structure may be made of a highly thermally conductive material, such as copper, aluminum or a ceramic, having a thermal conductivity of at least 50 W/mK. In particular when thermally coupled with electrically conductive layer structures of the stack, which may be made of copper and may thus have an excellent thermal conductivity as well, the reinforcing layer structure may form part of a heat removal path for removing heat generated in the component carrier during operation. For instance, a main heat source of such a component carrier may be an embedded electronic component, which may be thermally coupled with the reinforcing layer structure for heat dissipation.

[0032] In an embodiment, the reinforcing layer structure is electrically coupled with the at least one electrically conductive layer structure, for example for transmitting electric signals and/or electric power. Thus, the reinforcing layer structure itself may contribute to the electronic functionality of the component carrier.

[0033] In another embodiment, the reinforcing layer structure is functionally inactive apart from its reinforcing function. In particular, the reinforcing layer structure of such an embodiment may be decoupled electrically with respect to the electrically conductive layer structures of the stack. In terms of the actual functionality of the component carrier, the reinforcing layer structure of such an embodiment may be a dummy structure used only for reinforcing the inorganic layer structure during a manufacturing process.

[0034] In an embodiment, the reinforcing layer structure is electrically decoupled from the at least one electrically conductive layer structure. For instance, the reinforcing layer structure may be embodied as one or more metallic islands being separate from the other electrically conductive layer structures.

[0035] In an embodiment, the at least one inorganic layer structure forms a core of the stack. Hence, the inorganic layer structure may form a vertically central support body forming a base for additional organic layer structures on one or both opposing main surfaces thereof. Such a core structure may provide mechanical stability, may contribute to the electrical wiring of the component carrier (in particular for signal transport and/or supply of electric energy), and/or may contribute to heat removal.

[0036] In an embodiment, the at least one inorganic layer structure comprises or consists of at least one of glass, ceramic, semiconductor, for example silicon, and silica. Thus, appropriate materials for an inorganic layer structure are glass (in particular silicon-based glass), a ceramic (such as aluminum nitride and/or aluminum oxide), and a material comprising a semiconductor (such as silicon oxide, silicon, silicon carbide, gallium nitride, etc.). It is also possible to make the inorganic layer structure of a metallic material, such as copper. This may bring the advantage of having a big pool of material having individual chemical and/or physical properties and thus enable a freedom of material choice during design and/or manufacture of the stack.

[0037] In an embodiment, the reinforcing layer structure has a force sustain-ability of at least 40 Newton. With such a configuration, the reinforcing layer structure may be capable of withstanding forces as occur typically during a die attach process during which an electronic component is pressed into a cavity in the inorganic layer structure for attaching it.

[0038] In an embodiment, at least 50%, for example 60% to 80%, of a surface area of the second main surface overlapping with the surface area of the geometrically discontinuous region at the first main surface is covered by material of the reinforcing layer structure. This may ensure a sufficiently large reinforcing effect being achievable by the reinforcing layer structure.

[0039] In an embodiment, a first percentage of a coverage with material of the reinforcing layer structure in a first surface area of the second main surface overlapping with a surface area of the geometrically discontinuous region at the first main surface deviates (or differs) from a second percentage of a coverage with material of the at least one electrically conductive layer structure in a second surface area of the second main surface apart from the first surface area by not more than 20%, for example by not more than 10%. On the one hand, a certain deviation between a first

percentage and a second percentage may allow adjustment of the reinforcing function and the electric current conducting function of the component carrier individually. On the other hand, an excessive deviation between the first and second percentages may lead to a strongly asymmetric construction of the component carrier, which may increase warpage. When limiting the deviation to not more than 20% and preferably not more than 10%, a freedom of design and a low warpage configuration may be combined.

[0040] In an embodiment, a thickness of the at least one inorganic layer structure is not more than 500 μm , for example not more than 200 μm . In view of the continuing trend of miniaturization, a thin inorganic layer structure may be desired. However, very thin inorganic layer structures may render the latter prone to failure in the event of mechanical stress. By the provision of a reinforcing layer structure as described herein, stress exerted to the inorganic layer structure may be reduced, redirected, absorbed and/or suppressed so that the mentioned thin inorganic layer structures become possible without the risk of failure.

[0041] In an embodiment, the component carrier comprises at least one layer build-up which comprises organic material and which is formed on the first main surface and/or on the second main surface. For example, such a multilayer build-up may comprise an electrical wiring structure, a redistribution structure and/or one or more heat dissipating structures. One multilayer build-up may be formed on or above the inorganic layer structure by lamination, i.e. the application of heat and/or pressure. This may bring the advantage of forming complex electrically conductive layer structures in combination in an easy and well-established manner. However, said stack may also comprise one multilayer build-up on or above one main surface of the inorganic layer structure and/or the geometrically discontinuous region and another multi-layer build-up on or above another main surface of the inorganic layer structure and/or the reinforcing structure. In one embodiment, the multi-layer build-ups may be asymmetric with respect to each other. Although conventionally asymmetric multilayer build-ups on two opposing main surfaces of a core or another central carrier structure may be a root cause of warpage and delamination, the provision of a stiff and mechanically robust inorganic layer structure (preferably made of glass) in combination with a reinforcing layer structure also allows such an asymmetric build-up. This increases the freedom of design of the component carrier designer. Alternatively, the multilayer build-ups may be symmetric with respect to each other.

[0042] In an embodiment, a multilayer build-up on one main surface of the inorganic layer structure has a higher integration density of electrically conductive structures than another multilayer build-up on an opposing other main surface of the inorganic layer structure. The term “integration density” may denote a number of electrically conductive structures per area or volume of the respective region of the component carrier. In particular, the number of contacts (including pads) per area or volume of one multilayer build-up may be higher than the number of the contacts (including pads) per area or volume on the other multilayer build-up. Thus, integration density may mean a quantity of electrically conductive structures (such as traces) per mm^2 . The integration density above and below the inorganic layer structure can be different. For instance, integration density may be higher at a multilayer build-up facing a surface mounted device (such as a semiconductor chip). Thus, the multilayer build-up of the component carrier having the higher integration density may provide a mounting area for mounting one or a plurality of electronic components such as semiconductor chips. Preferably, the multilayer build-up of the component carrier having the higher integration density may be formed on the first main surface of the inorganic layer structure.

[0043] In an embodiment, the at least one electrically conductive layer structure comprises at least one vertical through connection extending over an entire thickness of the at least one inorganic layer structure. Advantageously, one or more vertical through connections such as copper pillars and/or copper filled laser vias extending through the entire thickness of the inorganic layer structure may electrically couple multilayer build-ups on two opposing main surfaces of the

inorganic layer structure. Thus, complex electrically conductive paths may be formed.

[0044] In an embodiment, the reinforcing layer structure comprises a matrix-like arrangement of reinforcing pads (see FIG. 8). Additionally or alternatively, the reinforcing layer structure comprises an array of reinforcing stripes arranged side-by-side (see FIG. 9). With such configurations, a simple manufacturing process and an effective reinforcement may be combined.

[0045] In an embodiment, the component carrier comprises a stack of at least one electrically insulating layer structure and at least one electrically conductive layer structure. For example, the component carrier may be a laminate of the mentioned electrically insulating layer structure(s) and electrically conductive layer structure(s), in particular layers formed by applying mechanical pressure and/or thermal energy. The mentioned stack may provide a plate-shaped component carrier capable of providing a large mounting surface for further components and being nevertheless very thin and compact.

[0046] In an embodiment, the component carrier is shaped as a plate. This contributes to the compact design, wherein the component carrier nevertheless provides a large basis for mounting components thereon. In particular a naked die as an example for an electronic component can be surface mounted on a thin plate such as a printed circuit board.

[0047] In an embodiment, the component carrier is configured as one of the group consisting of a printed circuit board, a substrate (in particular an IC substrate), and an interposer.

[0048] In the context of the present application, the term “printed circuit board” (PCB) may particularly denote a plate-shaped component carrier which is formed by laminating several electrically conductive layer structures with several electrically insulating layer structures, for instance by applying pressure and/or by the supply of thermal energy. As preferred materials for PCB technology, the electrically conductive layer structures are made of copper, whereas the electrically insulating layer structures may comprise resin and/or glass fibers, so-called prepreg or FR4 material. The various electrically conductive layer structures may be connected to one another in a desired way by forming holes through the laminate, for instance by laser drilling or mechanical drilling, and by partially or fully filling them with electrically conductive material (in particular copper), thereby forming vias or any other through-hole connections. The filled hole either connects the whole stack, (through-hole connections extending through several layers or the entire stack), or the filled hole connects at least two electrically conductive layers, called via. Similarly, optical interconnections can be formed through individual layers of the stack to receive and/or support an electro-optical circuit board (EOCB). A printed circuit board is usually configured for accommodating one or more components on one or both opposing surfaces of the plate-shaped printed circuit board. They may be connected to the respective main surface by soldering. A dielectric part of a PCB may be composed of resin with reinforcing fibers (such as glass fibers).

[0049] In an embodiment, the component carrier is an integrated circuit substrate. In the context of the present application, the term “integrated circuit substrate” (IC substrate) may particularly denote a component carrier having a size and a pitch adjusted to the requirements of an integrated circuit component (in particular a semiconductor chip) mounted thereon. An IC substrate may be a, in relation to a PCB, comparably small component carrier onto which one or more integrated circuit components may be mounted and that may act as a connection body between one or more chip(s) and a PCB or being plugged in a socket mounted on a PCB. For instance, an IC substrate may have substantially the same size as an electronic component to be mounted thereon (for instance in case of a Chip Scale Package (CSP)). In another embodiment, the IC substrate may be larger than the assigned component (for instance in a flip chip ball grid array, FCBGA, configuration). More specifically, an IC substrate can be understood as a carrier for electrical connections or electrical networks as well as a component carrier comparable to a printed circuit board (PCB), however with a considerably higher density of laterally and/or vertically arranged connections. Lateral connections are for example conductive paths, whereas vertical connections may be for example drill holes. These lateral and/or vertical connections are arranged within the IC

substrate and can be used to provide electrical, thermal, and/or mechanical connections of housed components or unhoused components (such as bare dies), particularly of IC chips, with a printed circuit board or an interposer. A dielectric part of an IC substrate may be composed of resin with reinforcing particles (such as reinforcing spheres, in particular glass spheres). A pitch, i.e. a distance between corresponding edges of two adjacent metal structures of an IC substrate may be not more than 150 μm , in particular not more than 100 μm . In contrast to this, a pitch of some PCBs may be at least 200 μm , in particular at least 300 μm .

[0050] The substrate or interposer may comprise or consist of at least a layer of glass, silicon (Si) and/or a photoimageable or dry-etchable organic material like epoxy-based build-up material (such as epoxy-based build-up film) or polymer compounds (which may or may not include photo- and/or thermosensitive molecules) like polyimide or polybenzoxazole.

[0051] In an embodiment, the at least one electrically insulating layer structure comprises at least one of the group consisting of a resin or a polymer, such as epoxy resin, cyanate ester resin, benzocyclobutene resin, Melamine derivatives, Polybenzoxabenzole (PBO), bismaleimide-triazine resin, polyphenylene derivative (e.g. based on polyphenylenether, PPE), polyimide (PI), polyamide (PA), liquid crystal polymer (LCP), polytetrafluoroethylene (PTFE), Bisbenzocyclobutene (BCB) and/or a combination thereof. Reinforcing layer structures such as webs, fibers, spheres or other kinds of filler particles, for example made of glass (multilayer glass) in order to form a composite, could be used as well. A semi-cured resin in combination with a reinforcing agent, e.g. fibers impregnated with the above-mentioned resins is called prepreg. These prepreps are often named after their properties e.g. FR4 or FR5, which describe their flame-retardant properties. Although prepreg particularly FR4 are usually preferred for rigid PCBs, other materials, in particular epoxy-based build-up materials (such as build-up films) or photoimageable dielectric materials, may be used as well. For high frequency applications, high-frequency materials such as polytetrafluoroethylene, liquid crystal polymer and/or cyanate ester resins, may be preferred. Besides these polymers, low temperature cofired ceramics (LTCC) or other low, very low or ultra-low DK materials may be applied in the component carrier as electrically insulating structures.

[0052] In an embodiment, the at least one electrically conductive layer structure comprises at least one of the group consisting of copper, aluminum, nickel, silver, gold, palladium, tungsten, titanium and magnesium. Although copper is usually preferred, other materials or coated versions thereof are possible as well, in particular materials coated with supra-conductive material or conductive polymers, such as graphene or poly(3,4-ethylenedioxythiophene) (PEDOT), respectively.

[0053] The at least one component can be selected from a group consisting of an electrically non-conductive inlay, an electrically conductive inlay (such as a metal inlay, preferably comprising copper or aluminum), a heat transfer unit (for example a heat pipe), a light guiding element (for example an optical waveguide or a light conductor connection), an electronic component, or combinations thereof. An inlay can be for instance a metal block, with or without an insulating material coating (IMS-inlay), which could be surface mounted for the purpose of facilitating heat dissipation. Suitable materials are defined according to their thermal conductivity, which should be at least 2 W/mK. Such materials are often based, but not limited to metals, metal-oxides and/or ceramics for instance copper, aluminum oxide (Al₂O₃) or aluminum nitride (AlN). In order to increase the heat exchange capacity, other geometries with increased surface area are frequently used as well. Furthermore, a component can be an active electronic component (having at least one p-n-junction implemented), a passive electronic component such as a resistor, an inductance, or capacitor, an electronic chip, a storage device (for instance a DRAM or another data memory), a filter, an integrated circuit (such as field-programmable gate array (FPGA), programmable array logic (PAL), generic array logic (GAL) and complex programmable logic devices (CPLDs)), a signal processing component, a power management component (such as a field-effect transistor (FET), metal-oxide-semiconductor field-effect transistor (MOSFET), complementary metal-oxide-semiconductor (CMOS), junction field-effect transistor (JFET), or

insulated-gate field-effect transistor (IGFET), all based on semiconductor materials such as silicon carbide (SiC), gallium arsenide (GaAs), gallium nitride (GaN), gallium oxide (Ga.sub.2O.sub.3), indium gallium arsenide (InGaAs), indium phosphide (InP), and/or any other suitable inorganic compound), an optoelectronic interface element, a light emitting diode, a photocoupler, a voltage converter (for example a DC/DC converter or an AC/DC converter), a cryptographic component, a transmitter and/or receiver, an electromechanical transducer, a sensor, an actuator, a microelectromechanical system (MEMS), a microprocessor, a capacitor, a resistor, an inductance, a battery, a switch, a camera, an antenna, a logic chip, and an energy harvesting unit. However, other components may be surface mounted on the component carrier. For example, a magnetic element can be used as a component. Such a magnetic element may be a permanent magnetic element (such as a ferromagnetic element, an antiferromagnetic element, a multiferroic element or a ferrimagnetic element, for instance a ferrite core) or may be a paramagnetic element. However, the component may also be an IC substrate, an interposer or a further component carrier, for example in a board-in-board configuration. The component may be surface mounted on the component carrier. Moreover, other components, in particular those components which generate and emit electromagnetic radiation and/or are sensitive with regard to electromagnetic radiation propagating from an environment, may be used as a component.

[0054] In an embodiment, the component carrier is a laminate-type component carrier. In such an embodiment, the component carrier is a compound of multiple layer structures which are stacked and connected together by applying a pressing force and/or heat.

[0055] After processing interior layer structures of the component carrier, it is possible to cover (in particular by lamination) one or both opposing main surfaces of the processed layer structures symmetrically or asymmetrically with one or more further electrically insulating layer structures and/or electrically conductive layer structures. In other words, a build-up may be continued until a desired number of layers is obtained.

[0056] After having completed formation of a stack of electrically insulating layer structures and electrically conductive layer structures, it is possible to proceed with a surface treatment of the obtained layers structures or component carrier.

[0057] In particular, an electrically insulating solder resist may be applied to one or both opposing main surfaces of the layer stack or component carrier in terms of surface treatment. For instance, it is possible to form such a solder resist on an entire main surface and to subsequently pattern the layer of solder resist to expose one or more electrically conductive surface portions which shall be used for electrically coupling the component carrier to an electronic periphery. The surface portions of the component carrier remaining covered with solder resist may be efficiently protected against oxidation or corrosion, in particular surface portions containing copper.

[0058] It is also possible to apply a surface finish selectively to exposed electrically conductive surface portions of the component carrier in terms of a surface treatment. Such a surface finish may be an electrically conductive cover material on exposed electrically conductive layer structures (such as pads, conductive tracks, etc., in particular comprising or consisting of copper) on a surface of a component carrier. If such exposed electrically conductive layer structures are left unprotected, then the exposed electrically conductive component carrier material (in particular copper) might oxidize, making the component carrier less reliable. A surface finish may then be formed for instance as an interface between a surface mounted component and the component carrier. The surface finish has the function to protect the exposed electrically conductive layer structures (in particular copper circuitry) and enable a joining process with one or more components, for instance by soldering. Examples for appropriate materials for a surface finish are Organic Solderability Preservative (OSP), Electroless Nickel Immersion Gold (ENIG), Electroless Nickel Immersion Palladium Immersion Gold (ENIPIG), gold (in particular hard gold), chemical tin, nickel-gold, nickel-palladium, etc.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0059] The aspects defined above, and further aspects of the present disclosure are apparent from the example embodiments to be described hereinafter and are explained with reference to these examples of embodiment.

[0060] FIG. 1 illustrates a cross-sectional view of a component carrier according to an exemplary embodiment of the disclosure.

[0061] FIG. 2 illustrates a cross-sectional view, and a plan view, of a component carrier according to an exemplary embodiment of the disclosure.

[0062] FIG. 3 illustrates a stress distribution at a planar surface of a component carrier according to an exemplary embodiment of the disclosure in comparison with a conventional component carrier, as obtained from simulations.

[0063] FIG. 4 illustrates a detail of a component carrier according to an exemplary embodiment of the disclosure.

[0064] FIG. 5 and FIG. 6 illustrate diagrams indicating results of a simulation of stress acting on a component carrier according to an exemplary embodiment of the disclosure in comparison with a conventional component carrier.

[0065] FIG. 7 illustrates a cross-sectional view of a component carrier according to an exemplary embodiment of the disclosure.

[0066] FIG. 8 illustrates a plan view of a component carrier according to an exemplary embodiment of the disclosure.

[0067] FIG. 9 illustrates a plan view of a component carrier according to an exemplary embodiment of the disclosure.

[0068] FIG. 10 illustrates a cross-sectional view of a component carrier according to another exemplary embodiment of the disclosure.

[0069] FIG. 11, FIG. 12, and FIG. 13 illustrate bottom main surfaces of a stack of a component carrier with a reinforcing layer structure according to different exemplary embodiments of the disclosure.

DETAILED DESCRIPTION OF ILLUSTRATED EMBODIMENTS

[0070] The illustrations in the drawings are schematically presented. In different drawings, similar or identical elements are provided with the same reference signs.

[0071] Before referring to the drawings, which illustrate exemplary embodiments, other aspects of the disclosure will be described in further detail, and some basic considerations will be summarized based on which exemplary embodiments of the disclosure have been developed.

[0072] Glass core substrates are considered as a replacement for organic substrates. In conventional approaches of manufacturing component carriers based on a glass plate with a cavity in which an electronic component (such as a die) is to be embedded, there may be issues concerning glass damage during die bonding. More specifically, a glass core may be prone to breakage during a die bonding and pressure manufacturing stage.

[0073] According to an exemplary embodiment of the disclosure, a component carrier (for instance a PCB) may be provided with a (for example laminated) layer stack having also one or more inorganic layer structures (which may be made of glass). Optionally, it may also be possible to have a further layer build-up including organic layer structures (such as prepreg or resin layers) on one or both opposing main surfaces of the inorganic layer structure. The inclusion of at least one inorganic layer structure may involve a potentially brittle structure tending to crack under the influence of mechanical stress, force or pressure. In particular when one of the main planar surfaces of an inorganic layer structure has a geometrically discontinuous region (for instance a recess for embedding a component), mechanical stress, pressure and/or force may be critical for the

mechanical integrity of the inorganic layer structure, in particular on its opposing other side. Beneficially, the other main surface of the inorganic layer structure opposing the geometrically discontinuous region may be equipped with a spatially partially or entirely corresponding reinforcing layer structure. Such a reinforcing layer structure may suppress mechanical stress caused by the geometrically discontinuous region when manufacturing the component carrier or during ordinary use. This may for instance reduce the danger of crack formation at the inorganic layer structure, for instance caused by pressing a component to be embedded into a cavity, such as a geometrically discontinuous region, of the inorganic layer structure. Forming the reinforcing layer structure as a thin layer structure may strictly limit space consumption of the component carrier in a vertical direction. As a result, a compact and reliable component carrier may be provided by an exemplary embodiment. In particular, exemplary embodiments of the disclosure may reduce the crack risk in a glass core cavity area and may reduce high stress deformation by the provision of a reinforcing layer structure.

[0074] A gist of an exemplary embodiment of the disclosure is a specific design for an inorganic layer stack with geometrically discontinuous region, in particular a glass core with cavity for embedding an electronic component therein. According to this design, a reinforcing layer structure in the form of a copper pad may be formed on the back side of the glass core in a cavity area to release stress exerted to the glass core when embedding the component. Thus, the provision of a reinforcing layer structure may lead to a mechanical stress depression. Advantageously, this may lower the risk of damage to the glass such as a crack. Furthermore, this may have a positive impact on warpage control of the component carrier. Thus, exemplary embodiments may bring the advantage of reducing process stress during manufacture to protect the glass, which may lower the risk of a glass crack. Beneficially, simulations of the die bonding process indicate the presence of less stress thanks to the provision of the reinforcing layer structure on the back side of the inorganic layer structure. After die bonding, there may be an encapsulation process for forming a layer build-up on the front side and/or on the back side of the at least one inorganic layer structure. For example, lamination of resin sheets or prepreg sheets may occur when forming a layer build-up. Lamination pressure may be exerted to the constituents of the component carrier in this context, and beneficially the reinforcing layer structure (for instance embodied as one or more simple copper pads) may release the stress and may thereby protect the at least one inorganic layer structure against breakage, crack or damage. In particular in view of larger and larger cavity sizes and/or numbers, larger and larger sizes and/or numbers of components to be embedded, thinner and thinner inorganic layer structure (such as one or more glass cores), the provision of a reinforcing layer structure (such as one or more properly designed and positioned copper pads) on the cavity area may make a panel or component carrier stronger and may lower the risk of crack formation during the manufacturing process. Simultaneously, a corresponding component carrier or panel may show improved warpage properties thanks to the provision of the reinforcing layer structure. Thus, the reinforcing layer structure may also balance warpage, especially in the presence of an asymmetric build-up or structure. Advantageously, a reinforcing layer structure embodied as one or more copper pads may render the copper density on the back side of the at least one inorganic layer structure of the panel or the component carrier more evenly balanced. This may lower the undulation risk of one or more laminated organic layer structures due to the volume of the dielectric layer for encapsulation in the cavity. The lamination on the two surfaces can be reduced, which may enable better control of the flatness of the dielectric layer(s).

[0075] When forming a component carrier with an inorganic layer structure, for instance in glass core technology, there may be a higher risk of glass crack issues compared with a purely organic substrate. Exemplary embodiments of the disclosure provide a design with a reinforcing layer structure on a main surface of an inorganic layer structure facing laterally a geometrically discontinuous region on the other main surface. This may lower the risk of glass crack issues and may contribute to a higher yield of a manufacturing process.

[0076] To put it briefly, an exemplary embodiment may apply a crack prevention portion, in form of the reinforcing layer structure, to a glass substrate with a cavity and an accommodated component. Thus, a crack of a glass substrate can be avoided. For example, the use of back side pads, as an example for the reinforcing layer structure, on the glass substrate may reliably avoid a crack of the glass substrate due to mechanical impact. In other words, one or more pads may be provided on the back side of an inorganic layer structure facing a geometrically discontinuous region, such as a cavity, on the front side for mechanical stress dissipation. Advantageously, a reinforcing layer structure embodied as one or more metal pads in a back side main surface region corresponding at least partially with an opposing geometrically discontinuous region in a front side main surface region can be manufactured simultaneously with other metal pads at the same height level but located peripherally. In other words, the reinforcing pads and the peripheral other pads may be manufactured in a common process, for instance by patterning a common metal layer. Thus, substantially no extra effort may be involved in the manufacture of the reinforcing layer structure.

[0077] According to an exemplary embodiment of the disclosure, a component carrier is provided which comprises a stack comprising at least one electrically conductive layer structure and at least one inorganic layer structure, said at least one inorganic layer structure comprising a first planar main surface and an opposed second planar main surface, wherein the first main surface comprises a disrupted portion (which may also be denoted as a geometrically discontinuous region) having a different extension from the main planar surface extension, and on the second main surface comprises a reinforcing layer structure at least partially overlapping the disrupted portion from the top view.

[0078] The following advantageous embodiments are possible. In an embodiment, the disrupted portion comprises a cavity. At least one component may be provided or embedded in the cavity, in particular a bridge die (such as a die having features such as exposed pads suitable to be connected to at least two components for connecting said at least two components). For example, an external electrically conductive layer structure on the first main surface and the component and/or the conductive area of the component are at a different vertical position. For instance, the disrupted portion comprises a protrusion extending beyond the first planar main surface. In an embodiment, the reinforcing layer structure is provided on the second main surface. For example, the reinforcing layer structure comprises a metallic layer structure, in particular corresponding to at least one portion of the at least one electrically conductive layer structure of the stack. In an embodiment, the at least one electrically conductive layer structure comprises a further portion of structures beside the reinforcing layer structure. In other words, the reinforcing portion and the further portion may be flush one to each other along the stack thickness direction. For instance, the reinforcing layer structure comprises traces and/or pads. For example, the reinforcing layer structure is in the form of a pattern of pads and/or traces. In an embodiment, the reinforcing layer comprises a continuous structure. In this context, “continuous” may mean not comprising patterned structures, for instance a foil shape. For example, the reinforcing layer structure has a planar extension within the planar extension of the disrupted portion. For instance, the reinforcing layer structure has a planar extension at least partially overlapping the boundary profile of the planar extension of the disrupted portion. In an embodiment, the reinforcing layer structure comprises at least a portion overlapping with one edge and/or corner portion of the disrupted portion. For example, the reinforcing layer structure comprises an inorganic layer structure (for instance comprising glass, a ceramic, etc.). In an embodiment, the reinforcing layer structure is configured and connected with the stack to dissipate heat and/or to be electrically connected to the at least one electrically conductive layer structure (for instance to transmit signals and/or for power transmission). For example, the reinforcing layer structure is insulated from the other electrically conductive layer structures. For instance, the at least one inorganic layer structure is a core for the stack. In an embodiment, the at least one inorganic layer structure comprises at least one of glass, ceramic, silicon, and quartz. For

instance, the force sustainability of the at least one inorganic layer structure and/or of the reinforcing layer structure is at least 40 N, for example is not more than 80 N. In an embodiment, the pressure distribution of the metallic structure depends on the design of the metallic structure. [0079] FIG. 1 illustrates a cross-sectional view of a component carrier **100** according to an exemplary embodiment of the disclosure. For instance, the component carrier **100** according to FIG. 1 may be embodied as an integrated circuit (IC) substrate or as a printed circuit board (PCB) with an inorganic core.

[0080] The illustrated component carrier **100** comprises a stack **102** having a plurality of electrically insulating layer structures **106**, **150** and a plurality of electrically conductive layer structures **104**. The electrically insulating layer structures **106**, **150** may comprise at least one inorganic layer structure **106** and at least one organic layer structure **150**. In the illustrated embodiment, stack **102** is composed of a lower sub-stack in form of a laminated lower multilayer build-up **122** and of a laminated upper sub-stack in form of an upper multilayer build-up **124**. A central sub-stack **152** may be a core between the multilayer build-ups **122**, **124**. Each of the layer build-ups **122**, **124** may comprise organic material, such as an epoxy resin, and may be formed on a respective one of a first main surface **108** and a second main surface **110** of the central sub-stack **152**. An adhesion promotor may be located between the central sub-stack **152** and the lower layer build-up **122** arranged on the second main surface **110** and/or an adhesion promotor may be located between the central sub-stack **152** and the upper layer build-up **124** arranged on the first main surface **108**.

[0081] Additionally, there may be a structure that forms a pattern directly on the glass surface (for instance formed by sputtering a metal film on the glass as seed layer). However, there might be a risk of delamination between the glass surface and seed layer, so an adhesion promotor may be added between the seed layer and the glass surface. The adhesion promotor can be a metal layer such as a titanium layer which can improve the adhesion between the glass and the seed layer. Or it may be possible to treat the glass surface by a chemical or gas to change the property of the glass surface to improve the adhesion between the glass and the seed layer or the resin.

[0082] The electrically conductive layer structures **104** may comprise patterned copper layers which may form horizontal pads and/or a horizontal wiring structure. Additionally or alternatively, the electrically conductive layer structures **104** may comprise vertical through connections such as copper pillars and/or copper filled laser vias. Moreover, the organic electrically insulating layer structures **150** may be for example prepreg or resin sheets. The outermost organic electrically insulating layer structures **150** may form a solder mask **154**. The central inorganic layer structure **106** is here embodied as a glass core or glass plate, but may also be made of a ceramic, a semiconductor, quartz, etc. For example, a thickness D of the inorganic layer structure **106** may be not more than 500 μm , for example 400 μm . Such very thin glass cores may be highly preferred for manufacturing compact component carriers **100** but may be prone to damage by a mechanical load. Alternatively, the thickness D of the inorganic layer structure **106** may be in a range from 500 μm to 1000 μm . The lower multilayer build-up **122** is formed below the inorganic layer structure **106** and the upper multilayer build-up **124** is formed above the inorganic layer structure **106**.

[0083] As shown, the inorganic layer structure **106** has a planar first main surface **108** and an opposing planar second main surface **110**. However, a geometrically discontinuous region **112** is formed in the first main surface **108**. The geometrically discontinuous region **112** is formed as a cavity **116** extending from the first main surface **108** into the inorganic layer structure **106**. For example, cavity **116** may be formed as a blind hole in the inorganic layer structure **106**. For instance, cavity **116** may be formed by laser processing, mechanical abrasion and/or etching. The cavity **116** may have a round shape, for example a cylindrical shape, or an edged shape, for example a squared shape. As shown, an electronic component **118** may be accommodated in the cavity **116** for embedding it in stack **102**. For instance, the electronic component **118** may be a semiconductor die, for example a bridge die. It is also possible that the electronic component **118** is

substituted by multiple electronic components in cavity **116**, or that a component with another functionality than an electronic functionality is inserted in cavity **116**. For example, a copper block or a ceramic block may be inserted as a component in the cavity **116** for heat removal purposes. Gaps or unfilled regions of the cavity **116** between the component **118** and the inorganic layer structure **106** may be filled by a filling material **156**. Filling material **156** may for instance be a resin material or a glue. For example, gaps or unfilled regions may be filled by laminating the organic layer structure **150** of layer build-up **124** on top of the inorganic layer structure **106** after embedding the component **118**. A resin flow from organic layer structure **150** into cavity **116** during lamination may then constitute the filling material **156**.

[0084] When inserting component **118** in cavity **116** formed in the planar main surface **108** of inorganic layer structure **106**, a tape **158** (such as a die bonding type) on a bottom main surface of the electronic component **118** may be pressed with pronounced force onto the bottom of the cavity **116** for fixing the component **118** in the cavity **116**. In view of the brittle and inelastic properties of glass or the like used for forming the inorganic layer structure **106**, crack issues may occur under the cavity **116** in conventional approaches, in particular in crack-critical regions shown schematically with reference sign **160** in FIG. **1**.

[0085] According to a preferred embodiment of the disclosure, the risk of damage of the inorganic layer structure **106** during the manufacturing process (in particular during an application of force or pressure for component embedding) may be reduced by providing a reinforcing layer structure **114** at the second main surface **110** of the inorganic layer structure **106**. For instance, the reinforcing layer structure **114** has a force sustainability of at least 40 Newton. As shown, the reinforcing layer structure **114** largely overlaps in a lateral direction with the geometrically discontinuous region **112** in the form of the cavity **116**. More precisely, in a viewing direction **174** perpendicular to parallel planar main surfaces **108**, **110**, the geometrically discontinuous region **112** deviating from planarity, on the one hand, and the reinforcing layer structure **114**, on the other hand, may be partially or even entirely in alignment with each other in a lateral direction, as indicated by alignment lines **162**. In FIG. **1**, the reinforcing layer structure **114** extends laterally over the extension of the geometrically discontinuous region **112** and is aligned at its exterior lateral ends with the exterior lateral ends of the geometrically discontinuous region **112**. Preferably, a laterally exterior outline of the geometrically discontinuous region **112** at the first main surface **108** may correspond to a laterally exterior outline of the reinforcing layer structure **114** on the second main surface **110** of the inorganic layer structure **106**. In an embodiment, an absolute value of a difference between an area delimited by an exterior outline of the reinforcing layer structure **114** and an area delimited by an exterior outline of the geometrically discontinuous region **112**, divided by the area delimited by the exterior outline of the geometrically discontinuous region **112** may be not more than 20%, preferably not more than 10%. Preferably, the reinforcing layer structure **114** may extend laterally over a boundary profile of the geometrically discontinuous region **112**, wherein said boundary profile may be formed by an edge and/or a corner portion of the geometrically discontinuous region **112**. Generally, the entire surface area or only part of the surface area of the second main surface **110** overlapping the surface area of the geometrically discontinuous region **112** on the first main surface **108** may be covered by reinforcing material, such as copper in the example of FIG. **1**. In the embodiment of FIG. **1**, about 60% to 70% of the surface area of the second main surface **110** overlapping the surface area of the geometrically discontinuous region **112** at the first main surface **108** is covered by material of the reinforcing layer structure **114**.

[0086] In the illustrated embodiment, the reinforcing layer structure **114** is constituted by an array of electrically conductive structures, such as copper pads, arranged on the second main surface **110** below the geometrically discontinuous region **112** at the first main surface **108**. Additionally or alternatively, the reinforcing layer structure **114** may comprise one or more electrically insulating elements, for example may form part of a correspondingly adapted second main surface **110** of the inorganic layer structure **106** (not shown).

[0087] Advantageously, the reinforcing layer structure **114** may form part of the electrically conductive layer structure **104** at the bottom side of the inorganic layer structure **106**, i.e. on the second main surface **110**. More specifically, the reinforcing layer structure **114** may be formed by a central part **164** of a patterned metal layer of said electrically conductive layer structure **104**. In addition, a peripheral portion **166** of said electrically conductive layer structure **104**, laterally surrounding said central part **164**, is formed apart from the reinforcing layer structure **114** and forms traces and/or pads of the component carrier **100**. Elements of said electrically conductive layer structure **104** of the peripheral portion **166** may thus be electrically connected with further electrically conductive layer structures **104** of the component carrier **100**. For instance, elements of said electrically conductive layer structure **104** of the peripheral portion **166** may be electrically coupled by vertical through connections **132** extending through the inorganic layer structure **106** up to electrically conductive layer structures **104** of the opposing layer build-up **124**. As shown, the vertical through connections **132** extend over the entire thickness D of the inorganic layer structure **106** and electrically interconnect the layer build-ups **122**, **124** with each other. Thus, elements of said electrically conductive layer structure of the peripheral portion **166** may have an electronic functionality.

[0088] In contrast to this, elements of said electrically conductive layer structure **104** of the central portion **164** constituting the reinforcing layer structure **114** may be non-functional apart from their reinforcing function. In particular, said elements of said electrically conductive layer structure **104** of the central portion **164** constituting the reinforcing structure **114** may have no electronic function during operation of the component carrier **100**. In such an embodiment, the reinforcing layer structure **114** may be functionally inactive apart from its reinforcing function and may be electrically decoupled from the other electrically conductive layer structures **104**. On the structure of the central portion **164** and/or the peripheral portion **166**, an adhesion promoter may be applied. This may reduce the risk of delamination.

[0089] In another embodiment, it may however be possible that the reinforcing layer structure **114** of the central portion **164** is a pattern which forms traces and/or pads having an electronic functionality during operation of the component carrier **100**, for instance being electrically coupled with other electrically conductive layer structures **104** of the stack **102**. The density or number of members per unit area of the reinforcing structure **114** in the central portion **164** may be different compared to the peripheral portion **166**. Alternatively, a density of or a number of members may be the same in the central portion **164** as in the peripheral portion **166**. Thus, the reinforcing layer structure **114** may also be electrically coupled with at least one of the electrically conductive layer structures **104** for transmitting electric signals and/or electric power during operation of the component carrier **100**.

[0090] In particular, a first percentage of a coverage with material of the reinforcing layer structure **114** in a first surface area of the second main surface **110** overlapping with a surface area of the geometrically discontinuous region **112** at the first main surface **108** may deviate or differ from a second percentage of a coverage with material of the at least one electrically conductive layer structure **104** in a second surface area of the second main surface **110** apart from the first surface area, for instance by less than 50%, preferably by 10%-20%. For example, the amount of copper used for reinforcing purposes in the central part **164** of the lowermost electrically conductive layer structure **104** may be selected differently from an amount of copper used for electrical connections purposes in the peripheral path **166** of the lowermost electrically conductive layer structure **104**. Besides the amount of copper, there may be different designs for the shapes of copper and/or a different size of the pattern. This enables adjustment of the copper content locally for optimizing the reinforcing and electric functions. However, the copper density below cavity **116** may not be too different from the copper density apart from the cavity **116** in order to avoid warpage resulting from an asymmetric configuration.

[0091] Still referring to multilayer build-up **122**, access holes **168** may be formed (for instance by

laser drilling or mechanically drilling) through electrically insulating layer structures **150** for exposing elements of the lowermost electrically conductive layer structure **104** belonging to the peripheral portion **166**. A surface finish **170** (for instance made of ENIG or ENEPIG) may be applied to exposed surfaces of said electrically conductive layer structure **104**. For example, the surface finish **170** may form an electric ground layer.

[0092] Now referring to multilayer build-up **124**, electrically conductive connection elements **172** may be applied on exposed surfaces of the uppermost electrically conductive layer structure **104**, which may also be exposed by forming access holes in the uppermost electrically insulating layer structure **150**. For example, said electrically conductive connection elements **172** may be solder bumps. Although not shown in FIG. **1**, one or more surface mounted components, for instance electronic components such as semiconductor chips, may be assembled on the electrically conductive connection elements **172**. Said surface mounted components may be coupled with each other via the embedded electronic component **118** in accordance with its functionality as bridge die. The embedded electronic component **118** may be externally accessible by copper laser vias and copper pads forming electrically conductive layer structures **104** above the electronic component **118**. As shown, an external electrically conductive layer structure **104** arranged externally from the inorganic layer structure **106** on the first main surface **108** thereof can be arranged at another vertical level than an electrically conductive pad **120** at an upper main surface of the component **118**.

[0093] FIG. **2** illustrates a cross-sectional view **176** and a plan view **178** of a component carrier **100** according to an exemplary embodiment of the disclosure.

[0094] The construction of FIG. **2** corresponds largely to the construction of FIG. **1** but only shows the central sub-stack **152** of FIG. **1**. Furthermore, FIG. **2** shows how pressure is applied by a pressure applicator **178** in a direction **180** during die attach of component **118** in cavity **116**. A support plate **182** (for instance made of a metal) on the bottom side of stack **102** holds the stack **102** in place during a die attach process. During this force application, the bottom side of the glass core may be prone to damage, and warpage **184** may occur in conventional approaches. However, by providing the reinforcing layer structure **114** on the second main surface **110** below the geometrically discontinuous region **112** on the first main surface **108** of the glass core, damage of the glass core may be avoided and warpage **184** may be reduced.

[0095] As shown in plan view **178**, the reinforcing structure **114** may be a matrix-like array of copper pads in a central part **164** of electrically conductive layer structure **104** on the second main surface **110** of the inorganic layer structure **106**.

[0096] FIG. **3** illustrates a stress distribution at a planar surface of a component carrier **100** according to an exemplary embodiment of the disclosure in comparison with a conventional component carrier, as obtained from simulations.

[0097] More specifically, FIG. **3** shows a stress distribution at a main surface **110** of an inorganic layer structure **106** without a reinforcing layer structure **114** (see upper row **192** in FIG. **3**) and with a reinforcing layer structure **114** (see lower row **194** in FIG. **3**). This comparison shows an applied die bonding force of 10 Newton (see left column **186** in FIG. **3**), 20 Newton (see middle column **188** in FIG. **3**), and 30 Newton (see right column **190** in FIG. **3**). As seen, the provision of reinforcing layer structure **114** significantly reduces the stress applied in particular to the central part of main surface **110**. More specifically, FIG. **3** shows out-of-plane deformation in millimeters after application of pressure in a die-bonding load case. The illustrated glass core stress analysis shows that the risk of failures in the glass core due to the die bonding force may be significantly reduced by the formation of the reinforcing layer structure **114** in regard to the stress/strain characteristics in the glass. According to FIG. **3**, the impact of the die bonding force is indicated by evaluating the stress in the glass core for the three mentioned bonding forces.

[0098] FIG. **4** illustrates a detail of a component carrier **100** according to an exemplary embodiment of the disclosure. More specifically, FIG. **4** shows critical regions **160** of the inorganic

layer structure **106** which are specifically prone to damage in the event of mechanical stress, and which can be reliably protected against failure by the application of the reinforcing layer structure **114**. Said critical regions **160** include a first position **160a** and a second position **160b**, which will be evaluated in further detail by FIG. 6.

[0099] FIG. 5 and FIG. 6 illustrate diagrams **200**, **210**, **220**, **230**, and **240** indicating results of a simulation of stress acting on a component carrier **100** according to an exemplary embodiment of the disclosure in comparison with a conventional component carrier. In particular, stress and strain results are shown in the glass-material.

[0100] Referring to diagram **200** of FIG. 5, three scenarios are shown for the application of a die bonding force of 10 Newton, 20 Newton, and 30 Newton. The results are plotted for a conventional component carrier without a reinforcing layer structure (see reference sign **202**), and a component carrier **100** according to an exemplary embodiment of the disclosure with a reinforcing layer structure **114** (see reference sign **204**). As shown in FIG. 5, the absolute deflection *d* (in mm) may be significantly reduced by the addition of the reinforcing layer structure **114**.

[0101] Now referring to FIG. 6, diagram **210** has an abscissa **212** plotting bonding force BF and has an ordinate **214** plotting compression stress CS for position **160a** of FIG. 4.

[0102] Diagram **220** has an abscissa **212** plotting bonding force BF and has an ordinate **224** plotting strain S for position **160a** of FIG. 4.

[0103] Diagram **230** has an abscissa **212** plotting bonding force BF and has an ordinate **234** plotting tensile stress TS for position **160b** of FIG. 4.

[0104] Diagram **240** has an abscissa **212** plotting bonding force BF and has an ordinate **244** plotting strain S for position **160b** of FIG. 4.

[0105] As shown in diagrams **210**, **220**, **230**, **240** the results are better in a scenario in which reinforcing layer structure **114** is present (see reference sign **204**) compared with a scenario in which reinforcing layer structure **114** is absent (see reference sign **202**).

[0106] FIG. 7 illustrates a cross-sectional view of a component carrier **100** according to an exemplary embodiment of the disclosure. As shown in FIG. 7, laminating additional organic layer structures **150** (such as resin sheets or prepreg sheets) on top and on bottom of the inorganic layer structure **106** by heated press plates **196**, **198** may also create mechanical stress. Also in this scenario, the reinforcing layer structure **114** opposing the geometrically discontinuous region **112** may protect the inorganic layer structure **106** against damage and may suppress warpage.

[0107] FIG. 8 illustrates a plan view of a component carrier **100** according to an exemplary embodiment of the disclosure. More specifically, FIG. 8 shows a plan view on main surface **110** of inorganic layer structure **106** being provided with a reinforcing layer structure **114**. In the illustrated embodiment, the reinforcing layer structure **114** comprises a matrix-like arrangement of reinforcing pads **128** in central part **164**, such as copper pads arranged in rows and columns.

[0108] FIG. 9 illustrates a plan view of a component carrier **100** according to an exemplary embodiment of the disclosure.

[0109] More specifically, FIG. 9 shows a plan view on main surface **110** of inorganic layer structure **106** being provided with a reinforcing layer structure **114**. In the illustrated embodiment, the reinforcing layer structure **114** comprises an array of parallel elongate reinforcing stripes **126** arranged side-by-side in central part **164**, such as copper strips arranged in parallel.

[0110] FIG. 10 illustrates a cross-sectional view of a component carrier **100** according to another exemplary embodiment of the disclosure.

[0111] The embodiment of FIG. 10 differs from the embodiment of FIG. 1 in particular in that, according to FIG. 10, the reinforcing layer structure **114** is formed as a continuous layer over the entire extension of an opposing geometrically discontinuous region **112**. Moreover, the geometrically discontinuous region **112** of FIG. 10 comprises a protrusion **111** extending upwardly from the first main surface **108** beyond the inorganic layer structure **106**. The protrusion **111** may be a ring surrounding a cavity **116** in which an electronic component **118** is inserted. In the

embodiment of FIG. 10, the reinforcing layer structure 114 is thermally coupled with the electronic component 118 by a vertical through connection in the form of an electrically conductive layer structure 104, which may be made of copper. Alternatively, a plurality of vertical through connections may connect the backside of the electronic component 118. This may enable the use of chips having a front side and back side connection. Thus, heat created by electronic component 118 during operation of the component carrier 100 may be removed via the vertical through connection and the reinforcing layer structure 114, so that the latter contributes for dissipating heat.

[0112] FIG. 11 to FIG. 13 illustrate bottom main surfaces 110 of a stack 102 of a component carrier 100 with a reinforcing layer structure 114 according to different exemplary embodiments of the disclosure.

[0113] Referring to FIG. 11, the reinforcing layer structure 114 comprises only one big pad covering the whole central part area. Hence, a single continuous structure may constitute the reinforcing layer structure 114 in the region opposing the geometrically discontinuous region 112 on the opposing side of the stack 102.

[0114] Referring to FIG. 12, the reinforcing layer structure 114 comprises an annular metallic structure. It is also possible that the reinforcing layer structure 114 comprises multiple (for example four) square pads surrounding multiple (for example four) edges of the central part. Thus, one or more rings may constitute the reinforcing layer structure 114 in the region opposing the geometrically discontinuous region 112 on the opposing side of the stack 102.

[0115] Referring to FIG. 13, the reinforcing layer structure 114 comprises multiple (for example four) pads at multiple (for example four) corners of a central part and/or at least one pad at the center of said multiple pads.

[0116] Hence, a plurality of centrally located and peripherally located pads may constitute the reinforcing layer structure 114 in the region opposing the geometrically discontinuous region 112 on the opposing side of the stack 102.

[0117] Generally, the reinforcing layer structure 114 may have a lot of different designs in accordance with a specific application for the sustainability of pressure required (for instance, a design of the reinforcing layer structure 114 can be calculated based on the required pressure sustainability for a certain design).

[0118] It should be noted that the term “comprising” does not exclude other elements or steps and the article “a” or “an” does not exclude a plurality. Also, elements described in association with different embodiments may be combined.

[0119] Implementation of the disclosure is not limited to the illustrated embodiments shown in the figures and as described above. Instead, a multiplicity of variants is possible which variants use the solutions shown and the principle according to the disclosure even in the case of fundamentally different embodiments.

Claims

1. A component carrier, comprising: a stack comprising at least one electrically conductive layer structure and at least one inorganic layer structure, wherein the at least one inorganic layer structure has a planar first main surface and an opposing planar second main surface; a geometrically discontinuous region at the planar first main surface; and a reinforcing layer structure at the opposing planar second main surface, wherein the reinforcing layer structure at least partially laterally overlaps with the geometrically discontinuous region.
2. The component carrier according to claim 1, wherein the geometrically discontinuous region comprises a cavity extending from the planar first main surface into the at least one inorganic layer structure.
3. The component carrier according to claim 2, wherein the component carrier comprises at least one component accommodated at least partially in the cavity.

- 4.** The component carrier according to claim 3, wherein an external electrically conductive layer structure of the at least one electrically conductive layer structure is arranged on the planar first main surface and is arranged at another vertical level than an electrically conductive pad at an upper main surface of the component.
- 5.** The component carrier according to claim 1, wherein the geometrically discontinuous region comprises a protrusion extending upwardly from the planar first main surface beyond the at least one inorganic layer structure.
- 6.** The component carrier according to claim 1, wherein the reinforcing layer structure is arranged on the opposing planar second main surface.
- 7.** The component carrier according to claim 1, wherein the reinforcing layer structure forms part of the at least one electrically conductive layer structure.
- 8.** The component carrier according to claim 1, wherein at least a portion of the electrically conductive layer structure is formed apart from the reinforcing layer structure.
- 9.** The component carrier according to claim 1, wherein the component carrier comprises at least one of the following features: the reinforcing layer structure forms traces and/or pads; the reinforcing layer structure is formed as a continuous structure; the reinforcing layer structure extends laterally over the entire geometrically discontinuous region; the reinforcing layer structure extends laterally over at least part of a boundary profile of the geometrically discontinuous region, wherein said boundary profile is formed by an edge and/or a corner portion of the geometrically discontinuous region; the reinforcing layer structure is inorganic and/or electrically insulating; the reinforcing layer structure is connected with the stack and configured to dissipate heat; the reinforcing layer structure is electrically coupled with the at least one electrically conductive layer structure for transmitting electric signals and/or electric power; the reinforcing layer structure is functionally inactive apart from its reinforcing function; the reinforcing layer structure is electrically decoupled from the at least one electrically conductive layer structure.
- 10.** The component carrier according to claim 1, wherein the at least one inorganic layer structure forms a core of the stack.
- 11.** The component carrier according to claim 1, wherein the at least one inorganic layer structure comprises at least one of glass, ceramic, semiconductor, silicon, and silica.
- 12.** The component carrier according to claim 1, wherein the reinforcing layer structure has a force sustainability of at least 40 Newton.
- 13.** The component carrier according to claim 1, wherein at least 50% of a surface area of the opposing planar second main surface overlapping a surface area of the geometrically discontinuous region at the planar first main surface is covered by material of the reinforcing layer structure.
- 14.** The component carrier according to claim 1, wherein a first percentage of a coverage with material of the reinforcing layer structure in a first surface area of the opposing planar second main surface overlapping with a surface area of the geometrically discontinuous region at the planar first main surface deviates from a second percentage of a coverage with material of the at least one electrically conductive layer structure in a second surface area of the opposing planar second main surface apart from the first surface area by not more than 20%.
- 15.** The component carrier according to claim 1, wherein a thickness of the at least one inorganic layer structure is not more than 500 μm .
- 16.** The component carrier according to claim 1, wherein the component carrier comprises at least one layer build-up which comprises organic material and which is formed on the planar first main surface and/or on the opposing planar second main surface.
- 17.** The component carrier according to claim 1, wherein the at least one electrically conductive layer structure comprises at least one vertical through connection extending over an entire thickness of the at least one inorganic layer structure.
- 18.** The component carrier according to claim 1, wherein the reinforcing layer structure comprises a matrix-like arrangement of reinforcing pads.

- 19.** The component carrier according to claim 1, wherein the reinforcing layer structure comprises an array of reinforcing stripes arranged side-by-side.
- 20.** A method of manufacturing a component carrier, comprising: providing a stack comprising at least one electrically conductive layer structure and at least one inorganic layer structure, wherein the at least one inorganic layer structure has a planar first main surface and an opposing planar second main surface; forming a geometrically discontinuous region at the planar first main surface; and forming a reinforcing layer structure at the opposing planar second main surface, wherein the reinforcing layer structure at least partially laterally overlaps with the geometrically discontinuous region.
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